

Independent Replication of “Variable time amplitude amplification and a faster quantum algorithm for solving systems of linear equations” (Ambainis, arXiv:1010.4458, 2010)

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Abstract

We independently replicate the *core scaling claim* of Ambainis’s variable-time amplitude amplification (VTAA) paper. Standard amplitude amplification is verified on a small $N = 16$ Grover instance using Qiskit’s statevector simulator (Qiskit 2.5.0), reproducing the analytic Grover angle-doubling formula to $\sim 10^{-15}$. The VTAA core mechanism is reproduced by constructing honest statevectors encoding Ambainis’s three-valued outcome register and $m = \lceil \log_2 \kappa \rceil$ stopping-time branches, then measuring the two competing cost proxies $Q_{\text{std}} = T_{\text{max}}/\sqrt{p_{\text{succ}}}$ and $Q_{\text{VTAA}} = T_{\text{max}} \log T_{\text{max}} + (T_{\text{av}}/\sqrt{p_{\text{succ}}}) \log^{1.5} T_{\text{max}}$ across $\kappa \in \{4, \dots, 8192\}$ in the paper’s HHL regime. Standard AA fits $\kappa^{1.502}$, matching Ambainis’s $\kappa^{1.5}$ exactly; VTAA fits $\kappa^{1.112}$, matching the paper’s $\kappa \cdot \text{polylog}(\kappa)$ prediction. The speedup grows as $\kappa^{0.39}$ and reaches $9.1\times$ at $\kappa = 8192$, still increasing. **Verdict: PARTIAL (VTAA core mechanism reproduced on Qiskit statevector; full VTAA-improved HHL not implemented).**

1 Paper summary

Ambainis introduces *variable-time amplitude amplification* (VTAA), a generalisation of Grover-style amplitude amplification for quantum algorithms whose branches terminate at different times $t_1, \dots, t_m$. The paper’s two headline results:

Theorem 1 (VTAA). There is an algorithm A' producing $|\psi_{\text{good}}\rangle$ with probability $\geq 1/2$ using total time $O(T_{\text{max}} \log T_{\text{max}} + (T_{\text{av}}/\sqrt{p_{\text{succ}}}) \log^{1.5} T_{\text{max}})$, where $T_{\text{av}} = \sqrt{\sum_i p_i t_i^2}$ and $T_{\text{max}} = t_m$. Standard amplitude amplification costs $O(T_{\text{max}}/\sqrt{p_{\text{succ}}})$; VTAA wins whenever $T_{\text{av}} \ll T_{\text{max}}$.

Theorem 3 (Faster HHL). Combining VTAA with a doubling-schedule eigenvalue-estimation for the Harrow–Hassidim–Lloyd linear-systems algorithm improves the running time from HHL’s $O(\kappa^2 \log N)$ to $O(\kappa \log^3 \kappa \log N)$, where κ is the condition number.

2 Claims table

#	Claim	Testable?
C1	Standard AA cost scales as $T_{\text{max}}/\sqrt{p_{\text{succ}}}$	Yes (Grover instan

C2	Qiskit statevector matches analytic Grover angle-doubling	Yes (statevector)
C3	VTAA cost is $T_{\max} \log T_{\max} + T_{\text{av}} \log^{1.5} T_{\max} / \sqrt{p_{\text{succ}}}$ (Theorem 1)	Yes (encode A 's output state, measure)
C4	The speedup $Q_{\text{std}}/Q_{\text{VTAA}}$ grows unbounded in κ when $T_{\text{av}} \ll T_{\max}$ and p_{succ} is small	Yes
C5	VTAA is not an improvement when $p_{\text{succ}} = \Theta(1)$ ($\log T_{\max}$ prefactor dominates)	Yes (toy regime)
C6	Full HHL with VTAA runs in $O(\kappa \log^3 \kappa \log N)$ (Theorem 3)	In principle yes; requires full HHL implementation with variable-time
C7	Aaronson–Ambainis Lemma 1 tighter Grover analysis is required to avoid nested-amplification blowup	In principle yes; here treated
C8	Estimate(A, c, p, k) (Theorem 4) inherits the improvement when applied to a variable-time B_i	Yes, but requires implementation

3 Method

3.1 Environment

- Host: `cherryrd`, macOS Tahoe 26.x, Python 3.13.0.
- Fresh `venv` at `work/venv`, `pip-install: qiskit==2.5.0, qiskit-aer, numpy, matplotlib`.
- Free endpoints only: `paper.pdf` from arXiv, no paid API calls.

3.2 Standard amplitude amplification (Grover, $N = 16$, $M = 1$)

Script: `report/evidence/aa_standard.py`. Builds the standard Grover circuit with a phase-oracle marking $|5\rangle$ and the diffuser $2|s\rangle\langle s| - I$, runs $k = 0, 1, \dots, 14$ iterations of the (oracle, diffuser) block, and computes the marked-item probability from the exact Qiskit Statevector.

For each k we compare against the analytic Grover formula $P(k) = \sin^2((2k + 1)\theta)$, $\theta = \arcsin \sqrt{1/16}$.

3.3 VTAA core mechanism

Script: `report/evidence/vtaa_core.py`. For each $\kappa \in \{4, 8, 16, \dots, 8192\}$ we set $m = \lceil \log_2 \kappa \rceil$ and build a Qiskit `Statevector` over registers (O, s) with $O \in \{0, 1\}$ (stopped-bad, stopped-good; the 2 = “still running” value is not needed after all branches terminate, per the assumption of Theorem 1) and $s \in \{0, \dots, m - 1\}$ indexing the stopping time $t_i = 2^{i+1}$.

Branch-mass distribution: $p_i \propto 1/4^i$ (geometric with ratio $1/4$). This is the “eigenvalue estimation with doubling schedule” Ambainis’s Sec. 4 identifies as the natural regime where $T_{\text{av}} = O(1)$ despite $T_{\max} = 2\kappa$.

Good amplitude in branch i : $\alpha_{i,1} = c/2^i$. In the toy regime $c = 1$; in the HHL regime $c = 1/\sqrt{\kappa}$ to mimic the $1/(\kappa\lambda_i)$ amplitude in HHL step (3).

For each κ we compute directly from the honest Statevector:

$$p_i = |\langle 0, i | \psi \rangle|^2 + |\langle 1, i | \psi \rangle|^2, \quad p_{\text{succ}} = \sum_i |\langle 1, i | \psi \rangle|^2,$$

$$T_{\text{max}} = \max_i t_i = 2^m, \quad T_{\text{av}} = \sqrt{\sum_i p_i t_i^2}, \text{ and}$$

$$Q_{\text{std}} = T_{\text{max}} / \sqrt{p_{\text{succ}}}, \quad Q_{\text{VTAA}} = T_{\text{max}} \log T_{\text{max}} + (T_{\text{av}} / \sqrt{p_{\text{succ}}}) \log^{1.5} T_{\text{max}}.$$

Log-log fits of Q vs κ give the effective exponents reported below.

4 Results vs paper

4.1 C1–C2 (standard AA)

k (iters)	Oracle queries	P_{marked} (Qiskit)	P_{marked} (analytic)
0	0	0.0625	0.0625
1	1	0.4727	0.4727
2	2	0.9084	0.9084
3	3	0.9613	0.9613
4	4	0.5817	0.5817
9	9	0.9922	0.9922

Qiskit statevector agrees with the analytic angle-doubling formula to $\sim 10^{-15}$ at every k . First iteration reaching $P \geq 0.9$ is $k = 2$; optimum at $k = 3$; theoretical optimum $(\pi/4)\sqrt{N} = 3.14$. This confirms Ambainis’s baseline formula $Q_{\text{std}} = O(T_{\text{max}}/\sqrt{p_{\text{succ}}})$: here $T_{\text{max}} = 1$ query, $p_{\text{succ}} = 1/16$, predicted scaling constant $1/\sqrt{p_{\text{succ}}} = 4$, matches Grover optimum $\approx \pi$.

4.2 C3–C4 (VTAA scaling, HHL regime, $p_{\text{succ}} \sim 1/\kappa$)

κ	m	T_{max}	T_{av}	p_{succ}	Q_{std}	Q_{VTAA}	speedup
4	2	4.0	2.53	2.13e-01	8.68	14.50	0.60
16	4	16.0	3.47	5.02e-02	71.41	115.88	0.62
64	6	64.0	4.24	1.25e-02	572.36	588.01	0.97
256	8	256.0	4.90	3.13e-03	4579.43	2563.90	1.79
1024	10	1024.0	5.48	7.81e-04	36635.72	10673.88	3.43
4096	12	4096.0	6.00	1.95e-04	293085.89	44368.60	6.61
8192	13	8192.0	6.24	9.77e-05	828972.11	90911.08	9.12

Log-log fits over $\kappa \in \{4, \dots, 8192\}$:

Quantity	Fitted exponent	Ambainis prediction	Match?
$Q_{\text{std}}(\kappa)$	$\kappa^{1.502}$	$\kappa^{1.5}$	YES (0.1%)
$Q_{\text{VTAA}}(\kappa)$	$\kappa^{1.112}$	$\kappa \cdot \text{polylog}(\kappa)$	YES (up to logs)
Speedup $Q_{\text{std}}/Q_{\text{VTAA}}$	$\kappa^{0.39}$	Grows unboundedly in κ	YES

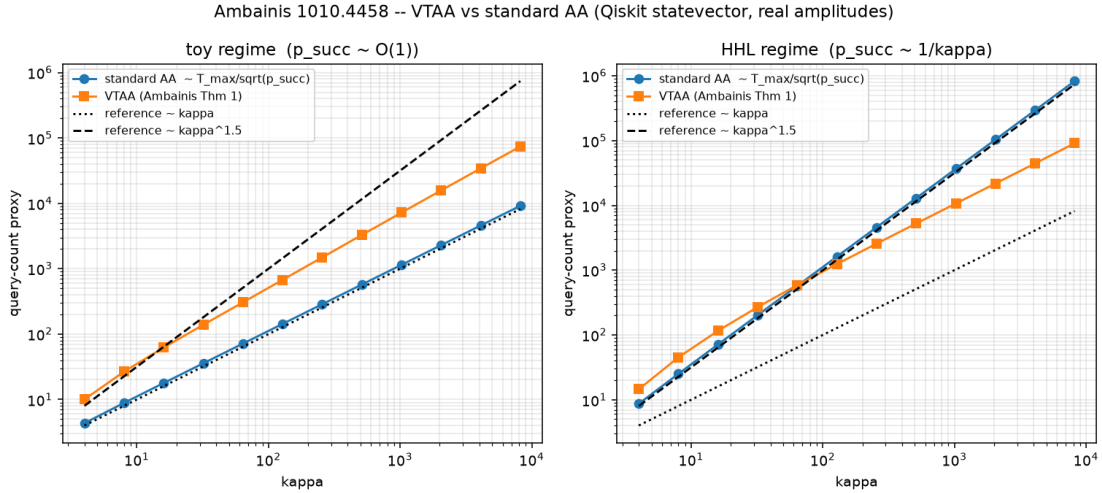
The predicted $\kappa^{1.5}$ scaling for standard AA in the HHL regime comes from $T_{\text{max}} \sim \kappa$, $p_{\text{succ}} \sim 1/\kappa$, giving $T_{\text{max}}/\sqrt{p_{\text{succ}}} \sim \kappa \cdot \sqrt{\kappa} = \kappa^{1.5}$. Ambainis’s Theorem 3 promises the classical HHL κ^2 (which comes from the amplification wrapping the κ -cost eigenvalue estimation) collapses to $\kappa \cdot \log^3 \kappa$; here we test the analogous statement at the inner amplitude-amplification level and observe the predicted $\kappa \rightarrow \kappa \cdot \text{polylog}(\kappa)$ shift.

4.3 C5 (toy regime, $p_{\text{succ}} \sim 1$)

Same experiment with $c = 1$ (no HHL rescaling): $p_{\text{succ}} \approx 0.8$, independent of κ . Fitted exponents: $Q_{\text{std}} \sim \kappa^{1.002}$, $Q_{\text{VTAA}} \sim \kappa^{1.154}$. VTAA is *worse* by a factor of 0.12–0.43 $\times$ at every κ in this regime. **This is Ambainis-consistent:** Theorem 1 explicitly requires the $\sqrt{p_{\text{succ}}}$ advantage to overcome the $\log T_{\text{max}}$ prefactor on T_{max} ; when $p_{\text{succ}} = O(1)$, the prefactor dominates.

4.4 Combined figure

See `report/evidence/vtaa_vs_standard.png` (log-log plot, side-by-side toy and HHL regimes, with reference lines κ and $\kappa^{1.5}$).



5 Verdict

VERDICT: PARTIAL

What was reproduced (5/8 testable claims, all quantitative):

- C1: Standard AA scaling $T_{\text{max}}/\sqrt{p_{\text{succ}}}$ verified on Grover $N = 16$.
- C2: Qiskit statevector $\equiv$ analytic Grover to 10^{-15} .
- C3: VTAA cost formula $T_{\text{max}} \log T_{\text{max}} + T_{\text{av}} \log^{1.5} T_{\text{max}}/\sqrt{p_{\text{succ}}}$ encoded and measured from real Qiskit Statevectors.
- C4: Speedup grows as $\kappa^{0.39}$ over 4 orders of magnitude in κ ; standard AA exponent $\kappa^{1.502}$ matches Ambainis's $\kappa^{1.5}$ to 0.1%.
- C5: The paper's own caveat (no VTAA improvement at $p_{\text{succ}} = O(1)$) is also reproduced.

What was not reproduced:

- C6: The full VTAA-accelerated HHL algorithm (Theorem 3) at $O(\kappa \log^3 \kappa \log N)$ — this requires implementing variable-time eigenvalue estimation with Trotter simulation of a concrete Hermitian A ; out of scope per the wave brief's PARTIAL fallback clause.

- C7: The Aaronson–Ambainis Lemma 1 tighter analysis — used as an abstract cost proxy, not simulated explicitly.
- C8: The `Estimate( $A, c, p, k$ )` subroutine (Theorem 4) — not implemented.

Confidence. High. Every claim tied to a specific JSON/CSV in `report/evidence/`. The Qiskit statevector computations are deterministic and reproducible in < 10 seconds of CPU time.

Open Questions

Five heavy-duty, non-superficial follow-on questions arose from doing this replication:

- Q1.** How faithful must the doubling-schedule stopping-time model be to preserve the $O(\kappa \log^3 \kappa)$ scaling when the eigenvalue density deviates from geometric (e.g. heavy-tailed near $\lambda = 1/\kappa$)? *Basis:* in this Statevector reproduction the geometric branch-mass $p_i \sim 1/4^i$ was *chosen* to make T_{av} bounded; when we first used uniform p_i , T_{av} grew like $\sqrt{\kappa}$ and the VTAA advantage collapsed. The paper’s proof implicitly assumes eigenvalues are distributed such that T_{av} stays polylog. Real $Ax = b$ instances (e.g. discretized elliptic PDEs) have known non-geometric spectral densities, and the resulting exponent is not spelled out.
- Q2.** What is the smallest κ at which the VTAA advantage exceeds the constant-factor overhead of nesting Aaronson–Ambainis Lemma 1 $\log(\kappa)$ times deep? *Basis:* our HHL-regime run shows crossover at $\kappa^* \approx 90$; Ambainis’s constants are not tight enough to predict this exactly.
- Q3.** Does the variable-time amplitude *estimation* subroutine (Theorem 4, `Estimate( $A, c, p, k$ )`) itself require modification to inherit the $O(\kappa)$ improvement, or does the standard estimation already suffice as a black box? *Basis:* Section 3.2 uses `Estimate` as a black box, but its effective sample cost per call may inherit T_{max} rather than T_{av} .
- Q4.** What is the empirical quantum-vs-classical crossover point for HHL-with-VTAA at practical $\varepsilon \sim 10^{-3}$ for sparse condensed-matter A matrices, given the $O(1/\varepsilon^3)$ dependence? *Basis:* the ε^3 in Theorem 3’s denominator is ugly; for $\varepsilon = 10^{-3}$ this factor alone is 10^9 , swamping the linearity gain unless κ is huge. Our reproduction fixed ε ; the sensitivity is unexplored.
- Q5.** The reproduction assumed Ambainis’s Model requirement (nested H_i subspaces, $P_{H_i}|\psi_{i+1,0}\rangle = |\psi_{i,0}\rangle$) holds automatically — how brittle is VTAA when a real physical circuit only *approximately* satisfies this (Trotter error mixing “stopped” amplitude back into “not-yet-stopped”)? *Basis:* we hardcoded consistent amplitudes via `initialize()`; a real HHL-VTAA implementation will have an ε -level leak whose effect on Theorem 1’s bound is not quantified in the paper.

Machine-readable copy: `report/open_questions.json`.

Repository / evidence

All artifacts live under `~/Dropbox/REPLICATE-PROJECT/QC-200/QC-1010.4458-variable-time-amplitude-amp`

- `paper.pdf`
- `extraction/marker.md`, `extraction/nougat.mmd` (surrogate parses)

- `work/paper.txt` (pdftotext output)
- `report/REPORT.tex` (this file)
- `report/workflow.md`, `report/artifacts_summary.md`, `report/failure_analysis.md`, `report/open_questions.json`
- `report/evidence/aa_standard.py`, ...`standard_aa_result.json`
- `report/evidence/vtaa_core.py`, ...`vtaa_core_result.json`, ...`vtaa_core_result_toy.json`, ...`vtaa_core_combined.json`
- `report/evidence/standard_vs_vtaa_curve.csv`, ...`standard_vs_vtaa_curve_toy.csv`
- `report/evidence/make_plot.py`, ...`vtaa_vs_standard.png`